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Software Engineering Group Project

Insert project title here

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Chapter 1

Introduction

Universities are large, complex organisations that provide a wide range of services to their students. At King's College London (KCL), students regularly need support across areas such as academic assessment, welfare, financial matters, careers, and university procedures. While specialist services exist for each of these areas, students frequently direct their queries to generalist staff—such as programme officers and personal tutors—who must then triage, redirect, or respond to a significant volume of incoming messages. This creates a recurring burden: staff spend considerable time handling queries that are routine, misdirected, or duplicated across multiple recipients, while overdue and unanswered queries risk going unnoticed altogether. The absence of a structured query-management process means that both staff efficiency and student experience suffer.

The KCL Ticketing System addresses this problem by providing programme officers with a centralised web application for managing incoming student queries as support tickets. Programme officers gain a single, structured workspace where every query is visible, trackable, and attributable—eliminating the risk of queries going unanswered and removing the duplicated effort that arises when the same query reaches multiple members of staff. Students benefit from a reliable, transparent channel for raising issues, with confidence that their query has been received and will be handled. The system further reduces staff workload by enabling immediate, automated preliminary responses to routine queries, reserving human attention for the cases that genuinely require it.

The system is a full-stack web application accessible through any modern browser. The back end is built with **Django 5** and **Django REST Framework** (Python), exposing a JSON API secured with JWT authentication. The front end is a single-page application built with **React 18** and **Redux Toolkit** (JavaScript). Email integration is provided by **Mailgun**, and the AI chatbot is powered by **Ollama** running a `llama2` model locally. Data is persisted in **SQLite** for development, with the architecture supporting migration to a production-grade relational database.

Chapter 2

Product

Instructions!

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Length requirements

There are no strict length requirements for this chapter. However, it is important that the writing is clear and concise. Avoid repeating yourself. It is expected that a typical project needs 2–4 pages, but this can vary considerably from project to project.

Expectations

This is a fairly short chapter that outlines what the team has developed and why. It will be used to try and understand the product your team has developed and how the features we can observe provide value to certain people. The chapter should describe what the team has delivered, not what the team hoped to deliver.
The section headers below provide a suggested structure. You should feel free to change it or extend it.

2.1 Objectives, context and key stakeholders

Universities such as King's College London are large, complex organisations that provide a wide range of services to students. Students frequently require support across areas including academic assessment, welfare, financial matters, careers, and general university procedures. While specialist services exist for each of these areas, generalist staff: such as programme officers and personal tutors, often act as the first point of contact. Personal tutors, who are typically teaching staff, must manage these queries alongside their existing teaching, research and administrative duties.

This situation creates several recurring challenges: programme officers and tutors spend substantial time handling misdirected or duplicated queries, while overdue or unanswered queries risk being overlooked. The lack of a structured query-management process reduces

staff efficiency and can negatively impact the student experience. For example, students may experience delayed responses to time-sensitive queries, and staff may encounter duplicated work or increased administrative burden.

2.1.1 Project Objectives

The primary objective of this project is to develop a Ticketing System that provides programme officers with a centralised web application for managing incoming student queries. The core aim is to create a single, structured workspace where every query can be viewed, tracked and is accounted for. The system aims to deliver the following benefits:

1. **Improve staff efficiency** – Reduce time spent on routine or duplicated queries, allowing staff to focus on complex or high-priority issues.
2. **Enhance student experience** – Offer students a reliable and transparent channel for submitting queries, with confirmation of receipt and updates on progress.
3. **Increase accountability and traceability** – Ensure that all queries are logged, assigned and tracked; preventing them from being lost or overlooked.
4. **Support evidence-based decision making** – Enable collection of data on query types, response times and resolution patterns to inform process improvements.
5. **Enable scalability** – Provide a system capable of supporting increasing query volumes.

2.1.2 Key Stakeholders

The main stakeholders of the Ticketing System are:

- **Programme officers and generalist staff** – Require a structured interface to manage queries efficiently, prioritise work and reduce repetitive tasks. They expect a reliable system that supports their workflow.
- **Students** – Seek timely, consistent support. They expect confirmation that their queries have been received and visibility of progress until resolution.
- **University administration** – Interested in improving operational efficiency and student satisfaction. They benefit indirectly from enhanced staff productivity and reporting capabilities that enable monitoring of service performance.

Secondary stakeholders include academic departments and IT support teams responsible for maintaining the system and ensuring smooth integration with existing university services.

2.1.3 Value Proposition

By centralising query management, the Ticketing System reduces duplicated effort, improves response times and provides a transparent communication channel for students. Staff can focus on cases that genuinely require human attention, while the university

gains actionable insights into recurring issues, supporting continuous improvement in student services. The system therefore, enhances efficiency, accountability and transparency; delivering measurable benefits to all stakeholders involved.

Expectations

This section aims to explain why you develop the project.
A good starting point are the project's key stakeholders and, insofar it is relevant, the background or context for the project. Provide enough information for the reader to understand the project objectives.
As you explain the project objectives, focus exclusively on what value the system adds to the stakeholders. Your objectives should justify *why* you are building the system. Do not confuse objectives with requirements: software requirements are *not* objectives. The software requirements explain *what* your system can do. In other words, they are a way to achieve objectives. Software requirements do *not* belong in this section. Indeed, if objectives and requirements are confused, the reader may be left with the impression that the project is a solution in search of a problem.
Your project should have at least one objective. Otherwise, the project is a pointless exercise. Having many objectives does not necessarily make for a better project, and could leave the project unfocussed.

2.2 Requirements and specifications

The requirements of the Ticketing System are derived directly from the project objectives described in Section 2.1. They define the capabilities and constraints of the system needed to meet the needs of students, programme officers and administrators. To present these requirements clearly, this section uses a user story approach, which expresses the system's functionality from the perspective of each stakeholder and links directly to the value the system provides.

Each user story describes a specific goal that a stakeholder wishes to achieve using the system. The corresponding system feature or page illustrates where the functionality can be experienced in the implemented platform. In addition to functional requirements, key non-functional requirements are listed to ensure performance, security, accessibility and scalability of the system.

The following tables summarise both the functional and non-functional requirements, providing a clear map between stakeholder needs, project objectives and system features:

Stakeholder	**User Story / Requirement**	**Objective Supported**	**System Feature / Page**
Student	Submit a support ticket with issue type, description and attachments	Enhance student experience	TicketFormPage
Student	View the status and history of submitted tickets	Increase accountability and transparency	UserDashboardPage
Student	Receive immediate confirmation or automated response upon ticket submission	Enhance student experience, improve staff efficiency	TicketFormPage, Automated Email / AI Chatbot
Student	Request meetings with staff, specifying date, time and topic	Improve staff efficiency, enhance student experience	MeetingRequestPage
Student	Close a ticket once their issue is resolved	Increase accountability and transparency	UserDashboardPage
Student	Reply to tickets after staff response	Enhance student experience	UserDashboardPage
Student	Receive progress updates on submitted tickets	Enhance student experience, improve transparency	UserDashboardPage
Staff / Programme Officer	View all tickets assigned to them	Improve staff efficiency	StaffDashboardPage
Staff / Programme Officer	Assign tickets to the appropriate staff member	Improve staff efficiency, increase accountability	StaffDashboardPage, TicketsManagement Component
Staff / Programme Officer	Update ticket status and close tickets once resolved	Increase accountability and transparency	StaffDashboardPage
Staff / Programme Officer	Accept or deny student meeting requests based on office hours	Improve staff efficiency	StaffDashboardPage, OfficeHours Management
Staff / Programme Officer	Redirect tickets to other staff members if more suitable	Improve staff efficiency, ensure correct support	StaffDashboardPage, TicketsManagement Component
Administrator	Create, update and deactivate user accounts with roles	Increase accountability and transparency	UsersManagement Component
Administrator	View, update or delete any ticket in the system	Increase accountability and transparency	TicketsManagement Component
Administrator	View average reply time per department	Support evidence-based decision making	Statistics Component / AdminDashboard
Administrator / Staff	Report inappropriate tickets	Maintain integrity of support	AdminDashboard/ StaffDashboardPage

Table 2.1: User story requirements mapped to system objectives and implemented features

ID	Non-Functional Requirement
NFR-1	The system shall be accessible via modern web browsers on both desktop and mobile devices.
NFR-2	The system shall authenticate users securely using JWT tokens.
NFR-3	Ticket submission and updates shall respond within 2 seconds under normal load.
NFR-4	The system shall maintain data integrity, ensuring that all tickets, replies, and meeting requests are reliably stored.

Table 2.2: Non-Functional Requirements

Expectations

Use this section to provide the requirements and/or specifications for the project. There are not strict rules on the format you should be using. You can employ user stories or more conventional descriptions of requirements/specifications. However, you should be consistent. Explain how the requirements/specifications derive from the objectives. If it might not be clear to the reader, identify where the reader can experience the implementation in the system that was delivered.

Chapter 3

Design

3.1 System Architecture

This section describes the overall architecture of the system and how the frontend, backend, and database interact to support the ticketing platform.

3.1.1 System Layers

The PackItUp ticketing system follows a three-tier client–server architecture consisting of a React frontend, a Django REST backend, and a relational database. This layered design separates presentation logic, application logic, and data storage.

The frontend is implemented as a React single-page application (SPA) running on port 3000, responsible for rendering the user interface, managing routing, and maintaining application state. The frontend communicates with the backend through RESTful API calls secured with JSON Web Token (JWT) authentication.

The backend is implemented using Django and Django REST Framework, running on port 8000, and provides the system's business logic including ticket management, meeting scheduling, and user authentication.

While an SQLite database is used for persistent storage during local development, the deployed production environment utilizes a cloud-hosted PostgreSQL database provided by Neon (neon.tech). Access to both is managed seamlessly through Django's Object Relational Mapping (ORM) layer, ensuring environment parity.

[System Architecture Diagram]

3.2 Component Design

This section describes the structure of the frontend components and how they organise the user interface and system functionality.

3.2.1 Frontend Pages

The frontend application is implemented as a React single-page application and is organised into modular page components located within the src/pages/ directory. Each page represents a major functional area of the system and interacts with the backend through REST API requests. For example, the UserDashboardPage allows students to view and manage their submitted tickets, while the StaffDashboardPage enables staff members to review and update assigned tickets. Ticket creation is handled through pages such as TicketFormPage, while additional pages provide supporting functionality including browsing staff members, requesting meetings, and viewing frequently asked questions.

3.2.2 Administrative Pages

Administrative functionality is implemented through specialised components located within the src/components/Admin/ directory. These components provide interfaces for managing system data and monitoring platform activity. The AdminDashboard acts as the central navigation hub for administrative operations and provides an overview of system statistics. Components such as TicketsManagement allow administrators to perform CRUD operations on tickets, including assigning staff members and updating ticket status. The UsersManagement component enables the management of user accounts and roles, while the Statistics component provides analytics and reporting capabilities, including the ability to export system data.

3.2.3 Shared Components

The application also includes shared components that provide reusable interface elements across multiple pages. These components ensure consistent navigation and layout throughout the system, for example the UserNavbar, which provides navigation for authenticated users, and components supporting the FAQ interface.

3.2.4 State Management and API Communication

Global application state is managed using Redux Toolkit, which provides a centralised store for managing shared data across the application. The authSlice manages user authentication state, while the adminSlice maintains state related to administrative dashboard data. Communication between the frontend and backend is handled through REST API calls using a centralised API utility (apiFetch). This utility automatically includes authentication credentials in requests by attaching a JSON Web Token to the Authorization header in the format Bearer <JWT_TOKEN>. Axios is also used for certain operations requiring additional request configuration.

[Frontent Component Diagram]

3.3 Backend Structure

This section describes the organisation of the backend application and the responsibilities of its main modules.

3.3.1 Django Applications

The backend is implemented using Django and Django REST Framework and is structured into several Django applications responsible for different areas of system functionality. The primary application, KCLTicketingSystems, contains the core logic of the ticketing platform, including ticket creation and management, user management, meeting request handling, and office hours management. An additional application, AIChatbot, provides an AI-powered support assistant using the Gemini API. This module offers a chatbot interface that allows users to interact with an automated support system through a Django-rendered chat page and REST API endpoints.

3.3.2 View Organisation

Backend functionality is implemented through Django REST views located within the KCLTicketingSystems/views/ directory. These views expose API endpoints responsible for key operations such as ticket creation, ticket updates, replies, staff dashboards, and meeting requests. Administrative functionality is handled through separate views that provide system management capabilities including user administration, ticket oversight, and reporting.

3.3.3 Permissions and Access Control

Access control within the backend is implemented using a role-based permissions system. Users are assigned roles such as student, staff member, or administrator, which determine the actions they are permitted to perform within the system. Most API endpoints require authentication through JSON Web Tokens, ensuring that only authorised users can access protected resources. Custom permission classes are implemented to restrict certain operations to staff or administrators, such as managing meeting requests or performing administrative actions.

3.3.4 AI Helper Development with Gemini API

An AI-powered support assistant was developed to enhance the student experience by providing immediate, automated responses to routine queries. This feature acts as a first-line support mechanism, potentially resolving issues before a formal ticket is created. The assistant is implemented using the Google Gemini API, integrated through a dedicated backend service to provide high-performance natural language processing.

From an architectural perspective, the AI helper follows a secure request-response pipeline.

Backend Proxying: The student submits a query through the React frontend to a protected Django endpoint, and the backend forwards the request to the Gemini API. This proxy-based design is a key security control because API credentials remain server-side and are never exposed in client-side code.

Prompt Engineering and Contextual Grounding: To keep responses relevant to King's College London (KCL) procedures, the system uses prompt engineering and contextual grounding. Queries are enriched with system-specific context, including academic assessment types, welfare services, and financial support workflows, to reduce hallucinations and improve policy-aligned, actionable output.

Stateless Inference: The integration uses Gemini's generative capabilities to interpret user intent and return sanitised JSON responses to the frontend. This supports a responsive, dashboard-integrated chat experience without requiring persistent model-side session state.

Operational safeguards were also implemented to maintain service quality.

Input Validation and Sanitization: User inputs are validated and sanitised at the Django layer before forwarding requests to Gemini, reducing prompt-injection risk and improving data integrity.

Graceful Error Handling: The backend includes exception handling for API failures such as rate-limit events and timeout errors. If a reliable answer cannot be generated, the interface guides the student to the TicketFormPage for manual support submission.

Optimization: Generation parameters, including temperature and maximum token limits, were calibrated to balance response depth with low-latency interactions in the student dashboard.

[Backend component diagram]

3.4 Database Design

[UML Class Diagram]

This section describes the structure of the system's persistent data storage and the relationships between the main entities used by the application.

The database schema is implemented using Django ORM models and is designed to support ticket management, communication between users and staff, and meeting scheduling. The central entity within the system is the User model, which extends Django's AbstractUser class to include additional attributes such as a KCL student identifier (k_number), department information, and a role field used to distinguish between student, staff, and administrator accounts.

Support requests are represented by the Ticket model, which stores information including the issue type, description, priority level, and current status of the ticket. Each ticket is associated with the user who created it and may be assigned to a staff member responsible for resolving the issue. Tickets may contain multiple replies and attachments, enabling threaded communication between users and staff members.

Replies are represented by the Reply model, which links users and tickets and allows users to respond to existing discussions. The model includes a self-referential relationship that supports nested replies, enabling threaded conversations within a ticket.

Meeting scheduling functionality is implemented through the MeetingRequest model. This entity stores meeting requests submitted by students to staff members and records details such as the requested meeting date and time, a description of the meeting topic, and the current request status. Staff availability is defined through the OfficeHours model, which records weekly time slots during which staff members are available to meet with students.

Attachments associated with support tickets are represented by the Attachment model, which stores file metadata and links uploaded files to the corresponding ticket.

Several constraints are enforced within the data model to maintain data integrity. User email addresses and student identifiers are required to be unique. Meeting requests are validated to ensure that the requested meeting time falls within the staff member's defined office hours, and office hour records enforce that start times occur before end times.

3.5 API Design

The frontend communicates with the backend through a RESTful API implemented using Django REST Framework. The API exposes endpoints for authentication, ticket management, meeting requests, and administrative operations. Most endpoints require authentication using JSON Web Tokens (JWT), which are included in the Authorization header of requests in the format Bearer <token>. API endpoints are organised according to user roles and system functionality.

3.5.1 Authentication Endpoints

Endpoint	**Method**	**Purpose**
/api/auth/register/	POST	Create a new user account
/api/auth/token/	POST	Obtain JWT access and refresh tokens
/api/auth/token/refresh/	POST	Refresh an expired access token
/api/users/me/	GET	Retrieve the currently authenticated user

Table 3.1: Authentication API endpoints

Authentication tokens are configured with a one-hour lifetime for access tokens and a longer validity period for refresh tokens.

3.5.2 Ticket Management Endpoints

Endpoint	**Method**	**Purpose**
/api/tickets/	POST	Create a new support ticket
/api/dashboard/	GET	Retrieve tickets belonging to the authenticated user
/api/dashboard/tickets/{id}/close/	POST	Close an existing ticket
/api/staff-dashboard/	GET	Retrieve tickets assigned to staff members
/api/staff/dashboard/{id}/update/	PATCH	Update ticket status

Table 3.2: Ticket management API endpoints

These endpoints allow users to create and manage support tickets while enabling staff members to review and update assigned tickets.

3.5.3 Meeting and Staff Endpoints

Endpoint	Method	Purpose
/api/staff/	GET	Retrieve list of staff members
/api/staff/{id}/	GET	Retrieve staff profile and office hou
/api/meeting-requests/	POST	Submit a meeting request
/api/staff/dashboard/meeting-requests/	GET	Staff view incoming meeting reque
/api/staff/dashboard/meeting-requests/{id}/accept/	POST	Accept a meeting request
/api/staff/dashboard/meeting-requests/{id}/deny/	POST	Reject a meeting request

Table 3.3: Meeting request and staff endpoints

Meeting requests are validated against staff office hours to ensure that appointments can only be scheduled during available time slots.

3.5.4 Administrative Endpoints

Endpoint	Method	Purpose
/api/admin/dashboard/stats/	GET	Retrieve system statistics
/api/admin/tickets/	GET	Retrieve all tickets
/api/admin/tickets/{id}/update/	PATCH	Update ticket information
/api/admin/tickets/{id}/delete/	DELETE	Delete a ticket
/api/admin/users/	GET	Retrieve all users

Table 3.4: Administrative API endpoints

3.6 System Workflows

This section illustrates the main operational processes of the system, such as ticket creation and meeting requests, and how different components interact to complete these tasks.

3.6.1 Ticket Creation Workflow

3.6.2 Meeting Request Workflow

3.6.3 Office Hours Management Workflow

3.7 Design Decisions

This section discusses the key architectural and technological choices made during development and the reasoning behind these decisions.

3.7.1 Choice of React and Django

The system was implemented using a React frontend and a Django REST backend. This combination was chosen to provide a clear separation between the presentation layer

and the application logic. React enables the development of a responsive single-page application that efficiently updates the user interface without requiring full page reloads, making it well suited to dashboard-style interfaces such as ticket management systems. Django REST Framework provides a structured and scalable backend environment for implementing business logic, data validation, and API endpoints.

Using a REST API between the frontend and backend also allows the system to support multiple clients in the future. For example, the same backend API could potentially be consumed by mobile applications or other services without requiring changes to the server-side implementation.

3.7.2 Benefits of the Architecture

The chosen architecture provides several advantages for the development and maintenance of the system. Separating the frontend and backend allows user interface development to be performed independently from server-side logic. React's component-based design supports reusable interface elements, while Django provides a structured backend environment with built-in tools such as the ORM and authentication framework. Together, these technologies enable a maintainable and scalable system architecture.

3.7.3 Limitations and Trade-offs

Despite its advantages, the chosen architecture introduces some trade-offs. Maintaining separate frontend and backend codebases increases development complexity compared to a single monolithic application. Additional configuration is also required to enable communication between the two layers, such as CORS and API proxy setup.

The system currently uses SQLite as the development database, which simplifies setup but is not suitable for large-scale production environments. However, the architecture allows migration to a more scalable database such as PostgreSQL if required.

The use of stateless JSON Web Token (JWT) authentication also introduces limitations, as tokens cannot easily be revoked before expiration without additional infrastructure. For the scope of this project, this trade-off was considered acceptable.

3.7.4 Alternative Approaches Considered

Several alternative approaches were considered during the design process. A server-rendered application using Django templates was evaluated but rejected due to its limited support for dynamic user interfaces. Other options, such as Next.js server-side rendering, GraphQL APIs, and microservice architectures, were also considered but were deemed unnecessarily complex for the scope of the project. A REST-based architecture using React and Django REST Framework provided a simpler and more appropriate solution.

Chapter 4

Testing

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Expectations

This chapter discusses the teams approach to software testing, as well as ensuring test quality. Cover the automated test suite included with the source code, as well as any less visible forms testing the team employed to ensure the marking criteria were met.

If your team used manual testing as well as automated testing, it is important to include the following:

- What did your rely on manual testing for? In other words, what was the purpose of manual testing.
- To what extent did you rely on manual testing for aspects of software testing that would ideally be automated.
- What are the respective limits of automated and manual testing.
- An appendix listing each manual test. For each manual test, describe precisely the actions the tester must undertake and what they are expected to observe. Also add when the test was last performed and whether it was successful.

Chapter 5

Project management

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Length requirements

There are no strict length requirements for this chapter. However, it is important that the writing is clear and concise. Avoid repeating yourself.

Expectations

This chapter is not assessed directly. Use it as an opportunity to reflect on how the project was managed, what worked well, what did not work so well, and what the team learned from the experience. If there is supporting information you wish to include (e.g. a team charter), add this in the form of appendices.

Project